Sahaj Bhadja

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Summary

Software engineer with 2+ years building secure APIs, real-time decision workflows, and AI-backed systems across Python, JavaScript/TypeScript, SQL, FastAPI, Flask, CI/CD, and fault-tolerant services.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured a fault-tolerant asynchronous orchestration service using FastAPI background workers, processing 1K+ AI jobs with explicit job-state transitions and failure isolation, improving throughput 3x under concurrent workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a production Flask backend on AWS serving 150+ instructors and 10K+ daily grading requests, with CI/CD-driven rubric workflows, secure file ingestion, and stable API performance under load.
- Designed a concurrent batch-processing pipeline using multithreading, OCR, and REST APIs to process 1K+ uploaded files per batch, maintaining fault-tolerant execution and clear debugging signals across grading workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for offline reinforcement learning in autonomous driving, improving long-horizon trajectory planning by 18% while developing end-to-end pipelines processing 100K+ trajectory logs reliably.
- Architected a spatial-temporal encoder that fused HD-map geometry with ego-agent context through multi-head attention, improving closed-loop planning stability by 15% across complex multi-agent merge and intersection scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with automated VM testing, executable builds, and 20+ unit and integration suites using Jenkins, PyTest, and Mockito, strengthening release reliability across three operating systems.
- Troubleshot large-scale API integrations with AWS Glue and S3, improving response times by 30% while leading ETL workflow improvements that boosted pipeline efficiency 68% and reduced technical debt 20%.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Designed a malicious URL detection system over 650K+ URLs, applying statistical feature engineering and machine learning models to classify phishing and malware patterns from behavioral indicators at scale.
- Achieved 96%+ detection accuracy using 21 URL behavior signals and deep learning architectures, generating actionable security insights into phishing campaigns, suspicious keywords, and infrastructure abuse trends globally.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Security & Platform: Identity security, Zero Trust, Conditional Access, policy engines, risk signals, agent identities, secure coding, authentication, authorization, row-level security, FastAPI, Flask, REST APIs, microservices, async workers
Cloud & Delivery: Azure SQL, PostgreSQL, SQL Server, AWS, S3, Docker, Jenkins, CI/CD, monitoring, debugging, distributed systems, ETL pipelines, LLM systems, PyTest, Git, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, Plaid*
- Built a scalable fintech platform for 50,000+ users and 1M+ transactions, combining Plaid, token-based firewall controls, CI/CD pipelines, and real-time updates to enforce secure transaction visibility reliably.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built secure document upload, summarization, and semantic retrieval workflows over 50K+ documents, using LLaMA-2 and FAISS to enable policy-aware search, collaborative review, and 95% query accuracy safely.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, PostgreSQL*
- Built an AI-powered SQL workspace supporting 10+ databases, combining Java and TypeScript services to accelerate schema exploration, query authoring, and structured execution across Windows, macOS, and Linux.